

Data Cleaning Report

Mobile Money Transaction Analysis — CSC 3221 Final Project

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Introduction

This report documents the data cleaning process performed on the mobile money transaction dataset. The goal was to transform the initial raw data into a structured, reliable, and analysis-ready format, ensuring accuracy and consistency for downstream analysis.

Raw Data Overview

The initial dataset (`raw_data_anonymized.csv`) contained transaction records with the following columns:

- UserId
- Date, Time
- Operator
- Transaction_type
- Direction
- Amount, Currency
- New_balance
- Anonymized_Content (full SMS message text)

Issues identified in the raw data:

- Inconsistent date/time formats
- Mixed currencies (XAF, FCFA)
- Duplicated or near-duplicate records
- Non-standardized transaction types and directions
- Embedded information in free-text fields
- Missing or anomalous values

Data Cleaning Steps

The following steps were applied to clean and standardize the data:

a. Standardization

- Unified date and time formats to ISO standard.
- Standardized currency representation (converted all to XAF where needed).
- Normalized transaction types and directions (e.g., 'paiement', 'transfert', 'retrait', 'transaction'; 'IN', 'OUT').

b. Deduplication

Removed exact and near-duplicate records based on UserId, Date, Time, Amount, and Transaction_type.

c. Handling Missing and Anomalous Values

- Imputed or removed records with missing critical fields (e.g., Amount, Date).
- Flagged and reviewed outliers in transaction amounts and balances.

d. Feature Extraction

- Parsed and extracted structured information from the Anonymized_Content field (e.g., sender/receiver, transaction IDs).
- Calculated derived features such as transaction frequency, average/median amounts, and balance statistics.

e. Anonymization

- Ensured all personal identifiers were replaced with anonymized tokens (e.g., [CONTACT_xxxx], [PHONE_xxxx]).

Cleaned Data Overview

The cleaned dataset (cleaned_data.csv) contains aggregated and feature-engineered records per user, with columns such as:

- UserId
- Total transactions, months active, tx per month
- Average, median, std, and total amounts (in/out)
- Transaction type counts and ratios (send/receive, weekend ratio)
- Balance statistics (average, min, max)
- Recency, velocity, and activity labels
- Demographic and socioeconomic features (age, gender, occupation, education, income, etc.)

Us erl d	total_tra nsaction s	month s_acti ve	avg_a moun t	total_a mount_i n	total_a mount_o ut	send_rec eive_rati o	avg_ balan ce	activi ty_lab el
US ER _1	147	30	9872. 82	779704. 0	671600.0	2.267	14101. 95	Low

Key Changes and Improvements

- Reduced noise and inconsistencies in transaction records.
- Enhanced data quality by removing duplicates and correcting errors.
- Added valuable features for analysis (e.g., transaction ratios, demographic links).
- Ensured privacy through robust anonymization.

Conclusion

The data cleaning process has resulted in a high-quality, anonymized, and analysis-ready dataset. This cleaned data forms the foundation for robust statistical analysis and modeling of mobile money usage patterns.